



FDM



HSBC

AI Hackathon Challenge

Internal team briefing

In financial services, specifications are critical for AI systems. They determine how models behave, establish guardrails for decision-making, and ensure regulatory compliance. Yet creating detailed specifications remains a bottleneck in the development process

That's why you'll begin by enhancing the provided business-specification agent—an AI tool that automatically generates specifications from business requirements. By improving this agent first, you're not just solving one problem; you're building the capability to solve dozens of future problems faster and more effectively.

Once enhanced, your improved agent will generate the specifications for your sustainable finance solution. Teams with better spec-generation capabilities will have clearer blueprints to work from, resulting in more focused solutions.

This approach mirrors how leading financial institutions are now working: using AI to accelerate the development of AI. It's a force multiplier that allows you to deliver higher quality solutions while meeting the stringent documentation requirements of the financial industry.

Schedule

Time	Topic	
9:30am to 10am	Hackathon Kickoff	- Welcome and partner introduction - Overview of AI SDLC and Spec-Driven-Development
10am to 12pm	Spec-Agent Enhancement and Review	- Refine the agent's ability to generate high-quality specifications from business requirements - Use their enhanced agent to generate specifications for respective sustainable finance use case
12pm to 1pm	Lunch	
1pm to 4pm	Solution Development	- Teams build their solutions according to specifications - Prepare presentation materials
4pm to 5pm	Final Demos and Presentations	- Final pitch and presentation to the judging panel - Q&A - Up to 15 minutes per team
5pm onwards	Networking Drinks and Refreshments	

Team 1	Use case 1
	Clean Power Ecosystem Intelligence
	HSBC Engineer – Michael Fok HSBC Engineer – Nicky Tan HSBC Engineer – Philip Yang FDM Consultant Benny Chung FDM Consultant Sannie Cheung FDM Consultant BA - Christy Tong

Team 3	Use case 3
	Solution Matching Engine
	HSBC Engineer – Pushpendra Bhurambe HSBC Engineer - Leong Wong HSBC Engineer - Bin Wang HSBC Engineer – Janice Zhang FDM Consultant Curtis Chang FDM Consultant Albert Mak FDM Consultant BA – Sam Chung

Team 2	Use case 2
	Sustainable Finance Alert System
	HSBC Engineer - Chanderler Qian HSBC Engineer - Elaine Mak HSBC Engineer - Nitin Sharma FDM Consultant Dennis Halim FDM Consultant Jonathan Chen FDM Consultant BA – Michelle Cheng

Team 4	Use case 4
	Cross-Ecosystem Opportunity Finder
	HSBC Engineer - Ashutosh Saxena HSBC Engineer - Nathan Zhu HSBC Engineer - Jayden Chau FDM Consultant Jemson Wong FDM Consultant Ryan Kwok FDM Consultant BA – Kevan Kenner

Challenge 1: Clean Power Ecosystem Intelligence

Business context:

HSBC has developed a strategic approach to the clean power sector that treats it as an interconnected ecosystem with eight key participant segments. As shown in the slides, we have mapped banking solutions to each segment, and we have successfully executed transactions across the ecosystem. Today, this ecosystem approach relies on manual mapping and analysis, which limits scale, consistency, and speed.

Problem statement:

How might we use technology, data, and AI to evolve our clean power ecosystem approach from a static framework into a dynamic, data-driven intelligence system that helps relationship managers identify, prioritise, and act on opportunities across all segments?

Key considerations:

- The ecosystem spans eight distinct segments, each with different banking needs.
- Relationships between entities across segments can create banking opportunities.
- Each segment requires specific banking solutions and expertise.
- Manual mapping cannot keep pace with the rapidly evolving clean power landscape.

Challenge 2: Sustainable Finance Alert System

Business context:

The clean power sector moves quickly. New projects, partnerships, regulations, and technology developments are announced daily. Many of these developments signal potential banking opportunities, but relationship managers cannot realistically monitor everything, or consistently translate updates into actionable outreach.

Problem statement:

How might we build an intelligent alert system that monitors developments in the clean power sector, identifies potential banking opportunities, and gives relationship managers timely, actionable insights?

Key considerations:

- Different news events create different banking opportunities.
- Each ecosystem segment has specific developments that matter most.
- The information volume is too high for manual processing.
- Timely outreach, backed by relevant insights, strengthens HSBC's position as a strategic adviser.

Challenge 3: Solution Matching Engine

Business context:

As shown in the slides, different participants in the clean power ecosystem require different banking solutions. Today, matching clients to the right solutions relies heavily on individual relationship manager expertise. That creates inconsistency, increases reliance on tacit knowledge, and can lead to missed opportunities.

Problem statement:

- How might we create an intelligent system that matches clients in the clean power ecosystem with the most appropriate banking solutions, based on their segment, characteristics, and needs?

Key considerations:

- Each ecosystem segment has specific banking solutions (as shown in the slides).
- Client characteristics within segments refine which solutions are most appropriate.
- The “right” solution can change as clients evolve within the ecosystem.
- Relationship managers need clear rationale for why a recommendation fits.

Challenge 4: Cross-Ecosystem Opportunity Finder

Business Context: Our clean power ecosystem framework recognizes that significant value is created at the intersections between different segments. For example, when Project Developers collaborate with EPC Contractors, or when Technology Suppliers connect with Storage Providers. These cross-segment relationships often represent prime banking opportunities, but our relationship managers typically specialize in specific segments rather than the entire ecosystem. Currently, we identify these cross-segment opportunities through individual RM knowledge and manual analysis, leading to inconsistent outcomes and missed opportunities.

Problem Statement: How might we create an intelligent solution that systematically identifies valuable cross-segment opportunities within the clean power ecosystem and equips our relationship managers with the insights needed to facilitate these connections and deliver appropriate banking solutions?

Key Considerations:

- Different segments in the clean power ecosystem have complementary banking needs and business relationships
- RMs typically specialize in specific segments rather than the whole ecosystem
- Cross-segment opportunities often go unnoticed due to organizational silos
- The most valuable banking opportunities frequently emerge at ecosystem intersections
- Successful cross-segment introductions can lead to multiple banking relationships and transactions
- Relationship managers need clear rationale and context when approaching clients about cross-segment opportunities

Tech Stack

Azure Cloud environment with Microsoft Foundry access :

- Claude: Opas 4.7
- GPT 5.4 mini, 4o, o4-mini, 4.1 mini
- DeepSeek V3.2
- Grok 4.2 - non reasoning

Virtual Machine environment on Azure with VS code pre-installed

- VSCode
- Ollama
- Docker Desktop/Docker Engine (issue with licensing for Docker Desktop)
- Node JS, npm
- Python
 - pandas
 - numpy
 - matplotlib
 - seaborn
 - langchain
 - langgraph
 - streamlit
 - FastAPI
 - Django
 - Tensorflow
 - Scikit-learn